

Computer Architecture Lab (Week 4)

Assembly Directive

- ❑ **What are assembler directives?**
- ❑ How do you set up an area of memory to be data?
 - For temporary data of the program
 - Set up constants for use in my program
 - And other possible uses
- **Assembler Directives are a means by which you can 'direct' the assembler to take specific actions.**

Assembly Directive

Assembly Directive	Syntax	Example	Description
@NEXT	@NEXT Memory_Address	@NEXT 4000	This assembler directive specifies where the next instruction is to be assembled in memory.
@BEGIN	@BEGIN Memory_Address	@BEGIN 2000	- Specify the memory location for execution begin - Can start anywhere in the program. - If there multi @BEGIN is used only the last one is considered.
@EQU	Label: @EQU Value	numA: @EQU 01	This assembler directive is used to declare constants. The symbol name is given as a label.

TP01: Calculate the sum of series of 8 bit numbers (no carry)

- ☐ **Statement:** Calculate the sum of series of 8-bits data from memory location 2000H to 2003H using branching instruction to loop. Memory location 2010H will be store the amount of series number to sum. Then store the sum data to address 2005H
- ☐ **Sample Problem:** 2000H = 05, 2001H = 04H
2002H = 50, 2003H = 10H
- ☐ **Result:** (2005) = 69H

LXI H 2000

LDA 2010

MOV B A

MVI A 00

NEXT: ADD M

INX H

DCR B

JNZ NEXT

STA 2005

HLT

TP02: Calculate the sum of series of 8 bit numbers (with carry)

- ☐ **Statement:** Calculate the sum of series of 8-bits data from memory location 2000H to 2003H using branching instruction to loop. Memory location 2010H will be store the amount of series number to sum. Then store the sum data to address 2005H
- ☐ **Sample Problem:** 2000H = 5A, 2001H = 10H
2002H = 74, 2003H = 85H
- ☐ **Result:** (2005) = 63H, (2006) = 01H

LXI H 2000
LDA 2010
MOV B A
MVI A 00
MVI C 00
NEXT: ADD M
JNC SKIP
INR C
SKIP: INX H
DCR B
JNZ NEXT
STA 2005
MOV A C
STA 2006

HLT

TP03: Calculate the sum of $1+2+...+n$

- ☐ **Statement:** Calculate the sum from 1 to n number in memory location 2000H.
Then store to address 2001H.
- ☐ **Sample Problem:** $(2000H) = 04H$
- ☐ **Result:** $(2001H) = 01 + 02 + 03 + 04 = 0A$


```
LDA 2000  
MOV B A  
MVI A 00  
MVI C 01  
NEXT: ADD C  
INR C  
DCR B  
JNZ NEXT  
STA 2001
```

```
HLT
```

**TP04: Write an assembly language to
calculate square of a number.**

MVI H 07
MVI L 07
XCHG
MOV C D
MVI D 00
LXI H 0000
LOOP: DAD D
DCR C
JNZ LOOP
SHLD 2001

HLT

TP05: Write an assembly language to calculate $(a+b)^2$ and store answer in location (2005)

```
LDA 2000  
MOV B A  
LDA 2001  
ADD B  
MOV B A  
MVI A 00  
MOV C B  
LOOP: ADD B  
DCR C  
JNZ LOOP  
STA 2005
```

```
HLT
```

TP06: Write an assembly language to calculate whether $m \times 2^n$ and store answer in address 2000

```
MVI A 05  
MVI C 03  
LOOP: RLC  
DCR C  
JNZ LOOP  
STA 2000
```

```
HLT
```

TP07: Write an assembly language to calculate $(a-b)^2$


```
MVI A 08  
MOV B A  
MVI A 03  
MOV C A  
MOV A B  
SUB C  
MOV D A  
MOV E A  
MVI A 00  
LOOP: ADD D  
DCR E  
JNZ LOOP  
STA 2000  
HLT
```

**TP08: Write an assembly language to
calculate whether $(a+b)^2$ or $(a-b)^2$**

MVI A, 01
MVI B, 08
MVI C, 03
CPI 01
JZ ADD
CPI 02
JZ SUB
ADD: MOV A, B
ADD C
MOV H, A
MOV L, A
MVI B, 00
MVI C, 00
JMP MULT
SUB: MOV A, B
SUB C

MOV H, A
MOV L, A
MVI B, 00
MVI C, 00
JMP MULT
MULT: XCHG
MOV C, D
MVI D 00
LXI H 0000
LOOP: DAD D
DCR C
JNZ LOOP
SHLD 2000

HLT

Thanks